

Quiz Master V2

(Quiz Master Version2 for MAD2 Project)

Student Details

- **Name:** Jatish Chawla
- **Roll Number:** 23f2000832
- **Course/Branch:** BS in Data Science

Problem Statement

Design and implement a full-stack web application for online quiz management. The system should allow admins to create/manage subjects, chapters, quizzes, questions, and users. Users can attempt quizzes, view their results, and track their analytics.

Approach to the Problem Statement

➤ Frameworks and Libraries Used:

- **Backend:**
 - Python 3.x
 - Flask
 - Flask-RESTful
 - Flask-JWT-Extended
 - Flask-CORS
 - SQLAlchemy (ORM)
 - SQLite (Database)
 - werkzeug.security (for password hashing)
- **Frontend:**
 - Vue.js 3
 - Vue Router
 - Axios
 - Chart.js (for analytics and charts)
 - Bootstrap 5 (for UI components)
 - HTML, CSS & JS
- **Redis** – Cachin and Celery task que backend
- **Celery** – Asynchronous task processing

➤ Approach:

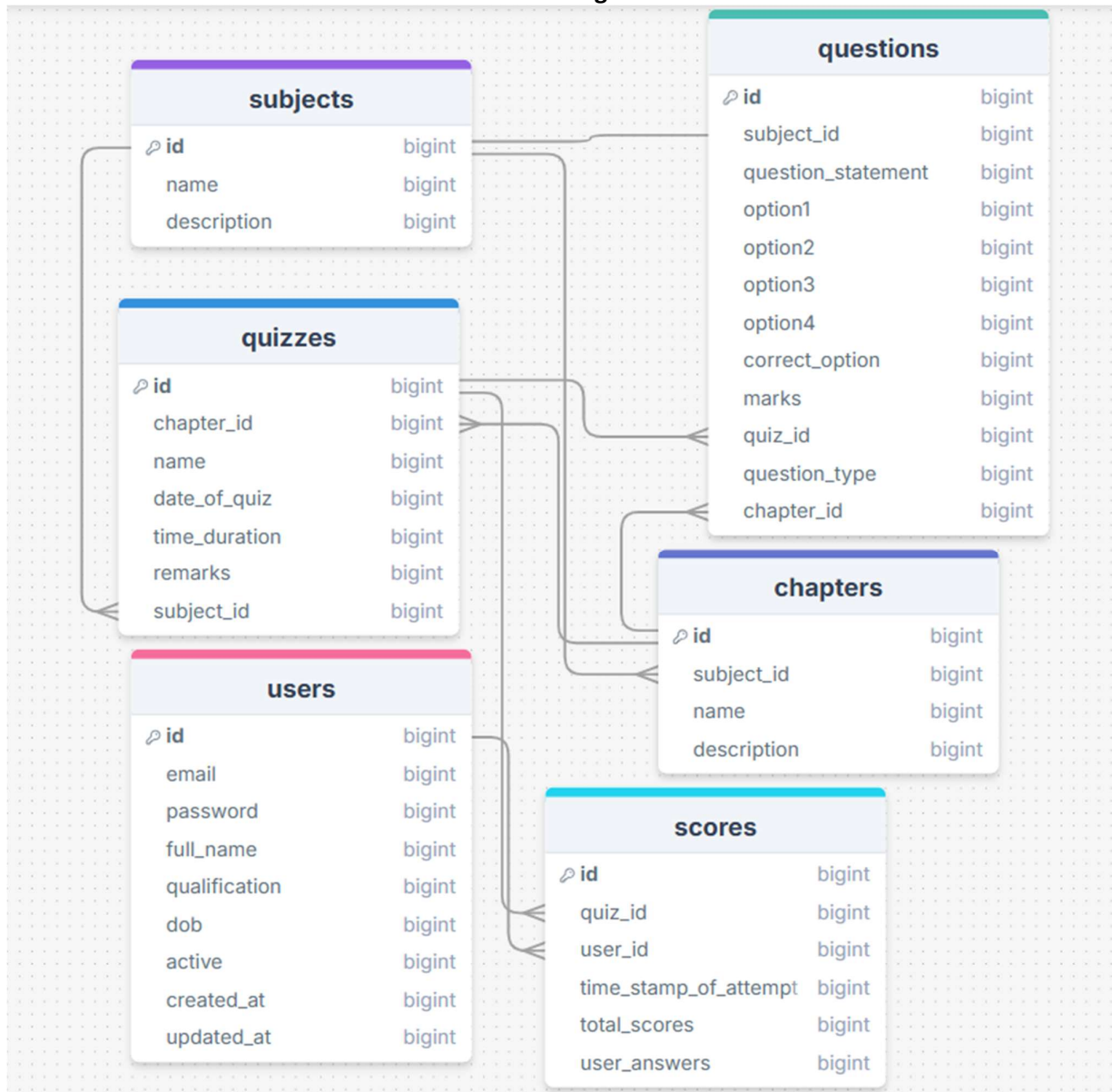
- Used a modular approach with a clear separation of backend (Flask) and frontend (Vue.js).
- Designed the database schema to support subjects, chapters, quizzes, questions, users, and scores.
- Implemented RESTful APIs for all CRUD operations and quiz attempts.
- Used JWT authentication for secure access and role-based authorization.
- Built a responsive frontend with Vue.js, supporting admin and user dashboards, analytics, and quiz interfaces.
- Added analytics and reporting features for both admins and users.

➤ Admin Capabilities in the Application:

- **Subject Management**
 1. Add, update, or delete subjects.
- **User Management**
 1. Manage customer and professional accounts and can view, edit and delete profiles.
 2. Block or unblock users based on ratings, reports or feedback.
- **Analytics Dashboard**
 1. Access dynamic charts showing statistics.
 2. Can view total Quizzes, Subjects, Chapters etc.
- **System Maintenance**
 1. Ensure platform integrity by overseeing key database entries and operational data.

➤ Designing the Database:

ER Diagram:



➤ Key Features

1. **Admin dashboard**: for managing subjects, chapters, quizzes, questions, and users
2. **User dashboard** with available quizzes, recent activity, and performance stats
3. **Quiz**: creation and management (CRUD) for admins
4. **Question**: creation and management (MCQ and text type)
5. Quiz attempt functionality for users with automatic scoring.
6. **User Authentication**: Implemented JWT token for authorization. Used hashing to store passwords. Secure login/registration for all users.
7. **RBAC**: Implemented Role Based Access Control
8. **Dynamic Charts**: Detailed analytics for both admins (site-wide) and users (personal progress)
9. Responsive frontend with modern UI (Bootstrap 5, Chart.js)
10. **Search**:
 - **Admin**: can search users/subjects/quizzes
 - **User**: can search subjects/quizzes by date/scores.

➤ **API Resource Endpoint:**

- /api/subjects [GET, POST], /api/subjects/<id> [GET, PUT, DELETE]
- /api/chapters [GET, POST], /api/chapters/<id> [GET, PUT, DELETE]
- /api/quizzes [GET, POST], /api/quizzes/<id> [GET, PUT, DELETE]
- /api/questions [GET, POST], /api/questions/<id> [GET, PUT, DELETE]
- /api/scores [GET], /api/scores/<id> [GET]
- /api/quizzes/<quiz_id>/attempt [POST]
- /api/subjects/<subject_id>/quizzes [GET]
- /api/quizzes/active [GET]
- /api/quizzes/expired [GET]
- /api/quizzes/<quiz_id>/details [GET]
- /api/users [GET, POST], /api/users/<id> [GET, PUT, DELETE]
- /api/profile [GET, PUT]
- /api/login [POST]

DRIVE LINK OF THE PRESENTATION VIDEO: